VIT
Vellore Institute of Technology

SLOT: A1+TA1+TAA1

SCHOOL OF ADVANCED SCIENCES
CONTINUOUS ASSESSMENT TEST – I
FALL SEMESTER 2025-26

Programme Name & Branch :B.Tech
Course Code :BMAT205L
Course Name : Discrete Mathematics and Graph theory
Faculty Name(s) : ANJANEYULU G S G N, PALLAVI MISHRA, ABHISHEK DAS PRAVEEN, SELVAKUMAR R, PARTHIBAN A, SUJASREE M
Class Number(s) : 2452,4418,0889,0869,0864,0893,0872
Date of Examination : 17.08.2025 (Session 1)
Exam Duration : 90 minutes Maximum Marks: 50

General instruction(s):

Answer All Questions

M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)

CO1. Learn proof techniques and concepts of inference theory
CO2. Use algebraic structures in applications

Q. No	Question	M	CO	BL
1.	Obtain the PDNF and PCNF of the following formula and hence conclude whether it is a tautology $(P \rightarrow (Q \wedge R)) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))$	10	1	BL3
2.	Test the consistency of the following Statements. i. If Jack studies well he will pass in exams. ii. If Jack studies well he will get a job. iii. Succeeding in exam and getting a job simultaneously are not possible for him. iv. Jack either enjoys or studies well. v. Finally, Jack enjoys.	10	1	BL3
3.	Show that the conclusion $(x)(F(x) \rightarrow \neg S(x))$ follows from (i) $(\exists x)(F(x) \wedge S(x)) \rightarrow (y)(M(y) \rightarrow W(y))$ and (ii) $(\exists y)(M(y) \wedge \neg W(y))$	10	1	BL3
4.	a) Prove that the set of all idempotent elements of a commutative monoid $(M, *)$ forms a submonoid b) Show that if every element in a group is its own inverse, then the group must be abelian.	5+5	2	BL2
5.	Prove that the order of a subgroup of a finite group divides the order of the group.	10	2	BL2

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